



# STANDARD OPERATING PROCEDURES

---

**MED**



# Table of Contents

|       |                                                                 |    |
|-------|-----------------------------------------------------------------|----|
| 1     | Hospital Treatment Protocol.....                                | 2  |
| 1.1   | Shelter Treatments Feline.....                                  | 3  |
| 1.1.1 | Feline URI.....                                                 | 3  |
| 1.1.2 | Feline Diarrhea.....                                            | 6  |
| 1.1.3 | Feline Skin Lesions.....                                        | 8  |
| 1.1.4 | Feline Wound Care.....                                          | 11 |
| 1.1.5 | Feline Ringworm (Dermatophytosis) Management.....               | 11 |
| 1.2   | Shelter Treatments Canine.....                                  | 13 |
| 1.2.1 | Canine URI.....                                                 | 13 |
| 1.2.2 | Canine Diarrhea.....                                            | 16 |
| 1.2.3 | Canine Skin Lesions.....                                        | 18 |
| 1.2.4 | Canine Wound Care.....                                          | 20 |
| 1.2.6 | Canine Diarrhea – Extended Treatment & Escalation Protocol..... | 21 |
| 1.3   | Perioperative (Unaltered) Procedures.....                       | 23 |
| 1.5   | Post-Operative Examinations.....                                | 25 |
| 1.5.1 | Post-OHE Examination.....                                       | 25 |
| 1.5.2 | Post-Castration Examination.....                                | 25 |
| 1.6   | Owned Animal Services.....                                      | 26 |
| 1.6.1 | Vaccination.....                                                | 26 |
| 1.6.2 | Microchipping.....                                              | 26 |
| 1.7   | Records Maintenance.....                                        | 26 |
| 2     | Controlled Substance Management.....                            | 27 |
| 2.1   | Locations/Contents.....                                         | 27 |
| 2.1.1 | Logs - Cabinet in Kitten Nursery (Main Shelter).....            | 27 |
| 2.1.2 | SAFE #3 – Floor Safe (Main Shelter).....                        | 27 |
| 2.1.3 | SAFE #4 – E-safe (Main Shelter).....                            | 27 |
| 2.1.4 | SAFE #1 – Medical Building - In Lab Area.....                   | 28 |
| 2.1.6 | SAFE #2 – Tech Safe – Transfer Safe (Medical Building).....     | 28 |
| 2.2   | Staff Access.....                                               | 28 |
| 2.3   | Procedures.....                                                 | 28 |
| 2.3.1 | Surgery – CVTs.....                                             | 28 |
| 2.3.2 | Dog/Cat Exams and Prepays by CVTs.....                          | 28 |

|       |                                 |    |
|-------|---------------------------------|----|
| 2.3.3 | Prepays by ACAs.....            | 28 |
| 2.3.4 | EUs by ACOs.....                | 29 |
| 2.4   | Ordering.....                   | 29 |
| 2.5   | Transferring Stock.....         | 29 |
| 2.6   | Inventories/Record Keeping..... | 30 |
| 2.6.1 | Daily.....                      | 30 |
| 2.6.2 | Weekly.....                     | 30 |
| 2.6.3 | Biennial.....                   | 30 |
| 2.8   | Theft/Loss Reporting.....       | 31 |
| 2.9   | Sample Forms.....               | 32 |

# 1 Hospital Treatment Protocol

**Purpose:** To provide standardized treatment protocols for use by Veterinary Technicians at Pinellas County Animal Services.

**Scope:** This SOP applies to all PCAS Veterinary Technicians authorized to administer medication and vaccines (under supervision of an attending veterinarian) to animals in residence at the shelter. Conditions that fall outside the parameters of this SOP will be treated in consultation with a PCAS Veterinarian.

**References:** Appropriate veterinary references.

## 1.1 Shelter Treatments Feline

### 1.1.1 Feline URI

Before initiating medication, always perform a complete physical exam. Treatment will be based on disease classification. Cats showing clear or colored nasal discharge, sneezing, clear or colored discharge from one or both eyes, conjunctivitis, ulcers/sores on nose, lips, or tongue, or has a fever, is lethargic or has a loss of appetite should be suspected of having URI and should be removed from the general population. Upper respiratory infection (URI) categories, standard and supportive treatments, and recommended diagnostic tests are included in this section.

#### Treatment Categories

| Category | Clinical Signs                                                                         | Probable Interpretation                                             | Treatment                                                                                                                                                                                     |
|----------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Clear/colored discharge (eye and/or nose), sneezing, squinting                         | Mild viral URI                                                      | Isolate in a separate room (Feline URI) or leave in place and note to clean and feed after healthy cats. Monitor appetite, hydration, and feces. Nebulize with saline if needed.              |
| 2        | Category 1<br>-AND-<br>More systemic signs (fever, inappetence, dehydration, lethargy) | Moderate to severe viral URI                                        | Category 1<br>-AND-<br>Monitor temperature, give SC fluids and nutritional support or supplementation if needed.                                                                              |
| 3        | Category 2<br>-AND-<br>Green, brown, yellow, bloody nasal or ocular discharge          | Viral URI with secondary non-specific secondary bacterial infection | Category 2<br>-AND-<br><b>Doxycycline</b> or Minocycline 10mg/kg q 24 hours until resolution of signs (10 days average). Re-evaluate at 3-5 days, consider alternate antibiotics. If there is |

|          |                                                                                                                                                                                                          |                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                                                                                                                          |                                                                        | <p>improvement but relapse after discontinuing antibiotic, consider testing for Chlamydia. If Chlamydia is confirmed or suspected, treat with doxycycline for 4-6 weeks (Azith ineffective). Okay for adoption when clinical signs resolve.</p> <p>If fails to respond to Doxycycline, give Enrofloxacin 5mg/kg q24h PO or SQ or <b>Azithromycin</b> 5-10mg/kg q 24h PO for 5 days the q 48h until resolution.</p> |
| <b>4</b> | <p>Category 3<br/>-AND-<br/>Green, brown, yellow, or bloody nasal discharge and/or moderate to severe conjunctivitis</p>                                                                                 | <p>Viral URI with moderate to severe secondary bacterial infection</p> | <p>Category 3<br/>-AND-<br/><b>Erythromycin Ophthalmic</b> OU BID x 7 days<br/>-OR-<br/>Tobramycin Ophthalmic OU BID x 7 days<br/>-AND-<br/>Flurbiprofen OU BID x 7 days<br/>-AND/OR-<br/>Antiviral Ophthalmics (Cidofovir 0.5%, Idoxuridine 0.1%, Ganciclovir 0.15%)</p>                                                                                                                                          |
| <b>5</b> | <p>Category 4<br/>-AND-<br/>Coughing, vomiting, diarrhea, swelling of any part of the body, rapid or difficulty breathing, fever &gt; 106<br/>-OR-<br/>Failure to respond to 2 rounds of antibiotics</p> | <p>Complicated URI or additional problems</p>                          | <p>Work up an individual plan with the veterinarian.<br/>Full veterinary Exam<br/>Supportive care as needed for dehydration, pyrexia, pain, congestion, anorexia (see Supportive Treatment)<br/>Perform testing (see Diagnostic Test)<br/>Rule-out additional medical problems</p>                                                                                                                                 |

## Supportive Treatments

| Condition   | Treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dehydration | <p>&lt;5% Subcutaneous fluids (<b>LRS</b> or Normosol-R) 250-500 ml q 12-24 hrs.<br/>           &gt;5% Administer intravenous fluids at 1-2 times the maintenance rate until dehydration is corrected</p> <p><b>Vitamin B12</b> or B Complex may be administered with fluids (1 ml per L). Cover fluid bag with a paper bag to prevent degradation of the vitamins.</p> <p>Add KCL to fluids for anorexic cats (20 mEq per liter)</p>                                                                                                                                                                                                                                                                                                                                               |
| Congestion  | <p>Clean nose with warm moist gauze</p> <p>Humidifier for 10 minutes q 6-24 hrs. (good for foster parents)</p> <p><b>Nebulization</b> for 10 minutes q 6-24 hrs. with sterile saline +/- albuterol, gentamicin, acetylcysteine</p> <p><b>Nose Drops</b> with Gentamicin +/- Dexamethasone – In each nostril BID as needed</p> <p>As a last resort <i>only if the cat is severely congested and not eating</i>, try a decongestant (phenylephrine). One drop BID before feeding in alternating nostrils. Left nostril AM and right nostril PM for no more than 3 days. Continuous use of decongestant nose drops will decrease its effectiveness and cause a rebound effect when stopped.</p>                                                                                        |
| Anorexia    | <p>Correct dehydration, treat congestion, provide analgesia, offer strong-smelling food. Consider appetite stimulants.</p> <p>Perform a complete oral exam. Severe ulceration can cause anorexia and will require analgesia.<br/>           If ulceration is observed, please check with veterinarian for possible outcome (euthanasia vs quarantine).</p> <p>If steps given above fail and the cat does not eat for &gt; 2 days, administer appetite-stimulating drugs: <b>Mirtazapine TD</b> 0.1ml or 15 mg tablets – 1/8 tablet q 24 hrs. PO (Mirtazapine demonstrates anti-nausea properties as well)<br/>           -OR-<br/>           Cyproheptadine 2–4 mg per cat q 12–24 hrs. PO<br/>           -OR-<br/>           Midazolam 1.25 mg SQ or IM once with food present</p> |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <p>If appetite does not respond to appetite stimulants, and anorexia persists for more than 5 days, consider placing an esophageal feeding tube.</p> <p>Monitor weight at least weekly</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Pain  | <p>Assess pain associated with oral or nasal planum ulceration, corneal ulceration, or severe systemic disease (e.g. joint pain associated with calicivirus, pain associated with dehydration), and treat accordingly. Assess renal function and hydration status prior to administration of NSAIDs.</p> <p><b>Buprenorphine SR</b> (sustained release) 0.12 mg/kg q72 hours SQ<br/> <b>Buprenorphine</b> 0.01 – 0.02 mg/kg q4-6 hrs. prn IM, SQ, trans-mucosal<br/> <b>Meloxicam</b> 0.1 mg/kg once PO, SQ. Subsequent dose starting 24 hours after initial dose at 0.05 mg/kg q 24 hrs. PO x 3 days.<sup>4</sup><br/> <b>Lidocaine</b> 4% viscous. Apply directly to affected area q 6-8 hrs. prn.</p> |
| Fever | <p>If temperature &gt; 106.5 F, administer fluids (subcutaneous or intravenous)</p> <p>Rule-out other systemic problems or lower respiratory tract involvement (thoracic radiographs if possible)</p> <p>Antipyretic drug treatment only if temperature &gt; 106.5 F and fails to respond to external cooling treatment</p>                                                                                                                                                                                                                                                                                                                                                                              |

## Diagnostic Tests

|                                 |                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------|
| FeLV / FIV Test                 | In-house ELISA tests to determine retrovirus status                                          |
| Polyp check                     | Full visual oro-nasopharyngeal examination under anesthesia                                  |
| PCR Respiratory Pathogen Panel  | Deep pharyngeal and/or nasal swab samples needed for submission to a laboratory (e.g. IDEXX) |
| Aerobic Culture and Sensitivity | Deep pharyngeal and/or nasal swab samples needed for submission to a laboratory (e.g. IDEXX) |
| Radiographs                     | Minimum of two views of thorax to determine lower respiratory tract involvement              |

### 1.1.2 Feline Diarrhea

| Assessment                               |                                                                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| History                                  | Testing                                                                                                                                       |
| Duration of illness                      | Fecal – Float                                                                                                                                 |
| Antibiotic history (past and/or present) | <p>Fecal – Direct (especially in chronic cases)</p> <p>Parvo SNAP test – perform if blood is present in stool, cat appears extremely ill,</p> |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydration status                        | or cat is found dead in kennel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Temperature                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Vomiting +/-                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Treatment</b>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Kittens                                 | <p><u>Symptomatic Support:</u><br/><b>Pedialyte</b> (unflavored/unsweetened) to replace water in formula<br/><b>Reduce formula concentration</b> to 1:3 parts powdered formula to water or Pedialyte<br/><b>Administer SQ fluids:</b> 5 ml/100 grams<br/><b>Fortiflora</b> - 1/3 packet added to each bottle of formula</p> <p><u>Medications:</u><br/><i>If suspect bacterial overgrowth:</i><br/>    <b>Clavamox</b> 0.05 ml - 0.1 ml BID PO for 5 days<br/>    -OR-<br/>    <b>Amoxicillin</b> 1 ml/ 10 lbs BID PO for 5 days<br/><i>For confirmed or suspected worms/giardia:</i><br/>    <b>Panacur</b> SID PO for 5 days<br/><i>For confirmed or suspected coccidia:</i><br/>    <b>Ponazuril</b> SID PO for 3 days</p> |
| Adults: Non-Complicated; Negative Fecal | <p><b>Administer SQ fluids:</b> 1 ml/ 10 lbs until dehydration corrected<br/>Begin treatment with <b>Metronidazole</b> 10-25 mg/kg SID PO for 5 days<br/>Feed bland diet for 3-5 days, then gradually reintroduce normal food</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Adults: Positive Fecal                  | <p><i>Roundworms/Hookworms</i> - <b>Strongid</b> 1 ml/10 lbs PO<br/><i>Coccidia</i> - <b>Ponazuril Paste</b> 1 ml/ 5 lbs + 21 cc water PO for 3 days<br/>    - OR -<br/>    <b>Ponazuril</b> (227 mg/ml suspension) - 1 ml/ 10 lbs PO for 3 days</p> <p><i>Giardia</i> - <b>Metronidazole</b> 10-25 mg/kg SID PO for 5 days<br/><i>Tapeworms</i> - <b>Praziquantel</b> (Droncit) 56.8 mg/mL injectable:<br/>    &lt;5lbs: 11.4 mg (0.2 mL)<br/>    5-10 lbs: 22.7 mg (0.4 mL)<br/>    &gt;10 lbs: 31.4 mg (0.6 mL maximum)</p>                                                                                                                                                                                                |
| Positive Parvo SNAP test                | <p>Single cat or kitten - euthanize<br/>Kitten with littermates:<br/><i>Option 1:</i> euthanize positive kitten and all littermates<br/><i>Option 2:</i> euthanize positive kitten → test all littermates with Parvo SNAP test<br/>    → move negative kittens to cat isolation in cat building → euthanize positive kittens (this option is dependent on the amount of space available in cat isolation; if there is no space available, consult with the staff veterinarian)</p>                                                                                                                                                                                                                                            |
| Chronic or                              | Consult staff veterinarian                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



|                                                        |  |
|--------------------------------------------------------|--|
| Complicated Cases<br>with Signs of Systemic<br>Disease |  |
|--------------------------------------------------------|--|

### 1.1.3 Feline Skin Lesions

| Things that cause crusting, redness & patchy hair loss |                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Skin condition                                         | Characteristics                                                                                                                                                               | Confirmatory Tests                                                                                                                                                                                                                                                                                                                                                                                                    |
| Dermatophytosis<br>(Ringworm)                          | Contagious and zoonotic *wear gloves*<br><br>Often pruritic<br><br>Several kittens of a litter will often be affected                                                         | <b>Positive Wood's lamp test</b><br>- This should NOT be used as a sole test; use in conjunction with skin scrape/fungal culture<br>- Urine, nasal discharge, medications, and fluorescein stain can glow under lamp<br><br><b>Positive DTM (traditional)</b> - media turns red within 10 days<br><br><b>Positive DTM (Rapid Vet-D)</b> - media turns red within 4 days<br><br><b>Response to therapy (see below)</b> |
| Sarcoptic Mange<br>(Scabies)                           | Contagious and zoonotic *wear gloves*<br><br>Severely pruritic                                                                                                                | <b>Positive skin scrape</b><br><br><b>Response to therapy (see below)</b>                                                                                                                                                                                                                                                                                                                                             |
| Pyoderma<br>(Bacterial Skin Infection)                 | Non-contagious<br><br>+/- pruritus                                                                                                                                            | <b>Skin cytology</b> <ul style="list-style-type: none"> <li>• impression smear</li> <li>• swabs of crusts and exudative skin</li> </ul> <b>Response to therapy (see below)</b>                                                                                                                                                                                                                                        |
| Allergic Dermatitis                                    | Non-contagious<br><br>Main Causes: fleas, food allergy, seasonal<br><br>History of pruritus (seasonal or nonseasonal)<br><br>Pruritus that persists beyond resolution of skin | <b>Rule out other causes</b> <ul style="list-style-type: none"> <li>• Negative skin scrape</li> <li>• Negative DTM</li> </ul><br>May see partial or complete response once starting proper <b>flea control</b> or after <b>Cytopoint</b> injection (see below for therapeutic guidelines)                                                                                                                             |

|         |                                                                                                                                                                                                             |                                                                           |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|         | infections/antibiotic therapy<br><br>Affected areas:<br><br>- Tail base & behind the ears<br>→ suspect flea allergy<br><br>- Involving ventrum +/- in between toes:<br>→ suspect food or seasonal allergies |                                                                           |
| Demodex | Not common<br><br>Most infections caused by D. canis; D. gatoi is usually extremely itchy and contagious                                                                                                    | <b>Positive skin scrape</b><br><br><b>Response to therapy (see below)</b> |

| Therapeutic Guidelines     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dermatophytosis (Ringworm) | Not recommended to house positive cats and kittens in shelter<br><br><i>Single spot</i> - <b>Miconazole 1% lotion</b> apply to skin SID or BID for at least 6 weeks based on culture results<br>DTM culture<br>Once 2 negative cultures are obtained (test weekly), treatment can be discontinued<br><br><i>Generalized, multiple spots:</i><br><b>Itraconazole</b> 5 mg/kg PO SID for 7 days on, then 7 days off (x 3 cycles)<br>DTM culture<br>Once 2 negative cultures are obtained (test weekly), treatment can be discontinued<br>-OR-<br><b>Lime Sulfur Dip</b> 2x weekly |
| Sarcoptic Mange (Scabies)  | <b>Bravecto</b> (topical): 0.07cc per lb<br>-OR-<br><br><b>Revolution</b> (topical): every 2 weeks; treat for 1-week past resolution of clinical signs<br><5 lbs: 0.13 ml<br>>5 lbs: 0.38 ml<br>-OR-<br><b>Ivermectin</b> (injectable) 0.01 mg/lb once weekly; treat for 1-week past resolution of clinical signs                                                                                                                                                                                                                                                               |
| Pyoderma                   | Surface Infection:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (Bacterial Skin Infection) | <p>Apply <b>2% to 4% chlorhexidine solution</b> topically to affected areas every 12 hours for 1-2 days past resolution of clinical signs<br/>( + topical ketoconazole if yeast present) - may use medicated shampoos</p> <p>Superficial Pyoderma:<br/> <i>*Treat for <u>at least 2 weeks, and 1-week past resolution of clinical signs</u></i><br/> <b>Medicated baths once weekly</b><br/> -AND-<br/> <b>Cefpodoxime</b> 5-10 mg/kg PO Q24h<br/> -OR-<br/> <b>Cephalexin</b> 22-30 mg/kg PO Q8-12h<br/> -OR-<br/> <b>Clavamox</b> 20 mg/kg PO Q8-12h</p> <p>Deep Pyoderma:<br/> <i>*Treat for <u>at least 4 weeks, and 1-week past resolution of clinical signs</u></i><br/> <b>Medicated baths once weekly</b><br/> -AND-<br/> <b>Cefpodoxime</b> 5-10 mg/kg PO Q24h<br/> -OR-<br/> <b>Cephalexin</b> 22-30 mg/kg PO Q8-12h<br/> -OR-<br/> <b>Clindamycin</b> 11 mg/kg PO Q12h<br/> *May be recommended when draining tracts, fistulas, or anal sac infections are also present<br/> -OR-<br/> <b>Clavamox</b> 20 mg/kg PO Q8-12h</p> |
| Allergic Dermatitis        | <p>Start on appropriate flea preventative - <b>Revolution</b><br/> For anti-inflammatory/ antipruritic effects:<br/> <b>Methylprednisolone</b> (Depo-Medrol) 5.5 mg/kg SQ or IM once<br/> -OR-<br/> <b>Prednisolone</b> start with 1 mg/kg PO BID for 3 days → then SID for 3 days → then SID every other day for 3 more doses</p> <p><i>*See treatment guidelines for pyoderma if secondary skin infections are present</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Demodex                    | <p>Lime Sulfur Dip 2x weekly<br/> -OR-<br/> <b>Bravecto Plus for Cats</b> - 1 dose (chewable or topical)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

*These are the most common feline skin lesions seen in a shelter setting. Other non-wound skin lesions such as tumors, etc. should be discussed with the staff veterinarian on a case-by-case basis*

#### 1.1.4 Feline Wound Care

- Assess wounds for depth, chronicity, and signs of infection
- Procedural sedation may be necessary to assess and treat wounds
- Clip and clean wounds whenever possible
- If maggots present: a Capstar tablet can be crushed and sprinkled on the wound bed. This will paralyze the maggots and make cleaning the wound easier.
- Superficial wounds can often be debrided with gauze then managed with **SSD, honey, or Zn7** topically BID until new skin has healed
- Flush deeper wounds (puncture holes, abscesses, etc.) with **diluted chlorhexidine solution**
- Some deeper wounds or abscess pockets may require surgery and placement of a drain by the staff veterinarian. Drains are usually left in place until the wound stops draining.
- For deep wounds, it is recommended to start antibiotic therapy to treat or prevent infections. Treatment duration is typically between 10 and 14 days, however it is recommended to continue antibiotic therapy for at least 2-3 days or 1-week past resolution of clinical signs.

Preferred antibiotics for cat wounds include:

- o **Convenia** - 3.6 mg/lb SQ q14 days
  - o **Clindamycin**
    - Capsules: 25 mg/ 10 lb BID
    - Liquid: 5 mg/lb SID
  - o Enrofloxacin (Baytril)
    - 5-20 mg/kg per day PO (SID or BID)
    - Young, pregnant, or suspect renal disease: 2.5 - 5 mg/kg PO or IM q12h
    - Maximum duration of treatment: 30 days
- 
- If wound infections fail to resolve or worsen after starting antibiotic therapy, it may be necessary to switch or add antibiotics to the treatment protocol. In rare cases, a culture and sensitivity may be indicated if the wound is not responding to any of the treatments listed above.

#### 1.1.5 Feline Ringworm (Dermatophytosis) Management

**Purpose:** To provide a consistent protocol for testing, treating, housing, and cleaning for kittens and cats suspected or confirmed to have ringworm, minimizing shelter exposure and promoting timely transfer to foster or rescue.

##### Testing

1. Any kitten/cat with skin lesions, hair loss, or other signs suspicious for ringworm should have a rapid dermatophyte (DTM) test performed – see 1.1.3 Feline Skin Lesions for confirmatory testing guidance (Wood's lamp, DTM).
2. While awaiting results, house the animal in the designated Observation Room.

### **Treatment While Awaiting Results**

Topical therapy is required for all animals pending results – either **Lime sulfur dip** twice weekly, or **Miconazole** topical application once daily. Record all treatment administration in the medical record.

### **If Test Is Positive**

1. Begin systemic antifungal treatment with **Itraconazole** 5 mg/kg PO SID for 7 days on, 7 days off (repeat cycles), per 1.1.3 Feline Skin Lesions therapeutic guidelines, dosed per veterinarian's instruction.
2. Topical therapy (lime sulfur dip or miconazole) may continue as an adjunct at the veterinarian's discretion.
3. Repeat DTM culture weekly until two consecutive negative results are obtained.
4. Do not house ringworm-positive animals long-term in the shelter – the medical team will prioritize placement into rescue or foster.

**After two negative repeat DTM tests:** discontinue antifungal treatment; the animal may move to regular adoption or foster pathways.

### **Housing**

All ringworm-suspect or -confirmed animals are housed temporarily in the Observation Room. Housing is intended to be short-term only; pursue foster or rescue placement as soon as possible.

### **PPE & Biosecurity**

1. Staff must wear gloves and a gown when handling animals in the Observation Room.
2. Use dedicated equipment (food/water bowls, litter pans, cleaning supplies) for the Observation Room – do not share with other areas.
3. Hand hygiene (soap and water) is mandatory after each handling; hand sanitizer alone does not kill ringworm spores (see 4.13 PPE & Hand Hygiene).

### **Cleaning Protocol**

1. Clean and disinfect all surfaces and equipment in the Observation Room daily.
2. Use **Rescue at 1:16** (8 oz/gallon) with a 10-minute contact time, or fresh **bleach at 1:10** (mixed daily) with a 10-minute contact time – see 4.1 Sanitation Standards & Products.
3. Always remove organic matter and pre-clean before applying disinfectant, and observe the full contact time.
4. Launder bedding/laundry from ringworm animals separately, using bleach.

### **Documentation**

Record all test results, treatments, and housing notes in the medical record, including DTM collection dates and results, to guide timing of repeat testing.

## 1.2 Shelter Treatments Canine

### 1.2.1 Canine URI

Before initiating medication, always perform a complete physical exam. Treatment will be based on disease classification. Dogs showing clear or colored nasal discharge, snorting, sneezing, a dry or productive cough, clear or colored discharge from one or both eyes, or has a fever, is lethargic or has a loss of appetite should be suspected of having URI and should be removed from the general population. Upper respiratory infection (URI) categories, standard and supportive treatments, and recommended diagnostic tests are included in this section.

#### Treatment Categories

| Category | Clinical Signs                                                                         | Probable Interpretation                                             | Treatment                                                                                                                                                                                                                                                                                                                                 |
|----------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Clear/colored discharge (eye and/or nose), sneezing, squinting                         | Mild viral URI                                                      |                                                                                                                                                                                                                                                                                                                                           |
| 2        | Category 1<br>-AND-<br>More systemic signs (fever, inappetence, dehydration, lethargy) | Moderate to severe viral URI                                        | Category 1<br>-AND-<br>Monitor temperature, give SC fluids and nutritional support or supplementation if needed.                                                                                                                                                                                                                          |
| 3        | Category 2<br>-AND-<br>Green, brown, yellow, bloody nasal or ocular discharge          | Viral URI with secondary non-specific secondary bacterial infection | Category 2<br>-AND-<br><b>Doxycycline</b> or Minocycline 10mg/kg q 24 hours until resolution of signs (14 days average). Re-evaluate at 7 days, consider alternate antibiotics such as <b>Enrofloxacin</b> 5mg/kg q24h PO or SQ with <b>Clindamycin</b> or <b>Azithromycin</b> 5-10mg/kg q 24h PO for 5 days then q 48h until resolution. |
| 4        | Category 3<br>-AND (any of the following)                                              | Severe URI -prolonged or possibly progressed to pneumonia           | Coughing – <b>Cerenia</b> EOD – 2mg/kg or <b>Butorphanol</b> (0.05–0.1 mg/kg) IM/SQ q4–6h. 10mg/mL)                                                                                                                                                                                                                                       |

|  |                                                           |  |                                                                                                                                                                         |
|--|-----------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Coughing, cheek puffing, lethargy with signs of pneumonia |  | Cheek puffing/poor oxygenation –<br><b>Pentoxiphylline</b> - 10–15 mg/kg 12h.<br>Pneumonia dose of <b>Enrofloxacin</b> – 10mg/kg SID and <b>Clindamycin</b> 10mg/kg BID |
|--|-----------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Supportive Treatments

| Condition   | Treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dehydration | <p>&lt;5% Subcutaneous fluids (<b>LRS</b> or Normosol-R) 250-500 ml q 12-24 hrs.</p> <p>&gt;5% Administer intravenous fluids at 1-2 times the maintenance rate until dehydration is corrected</p> <p><b>Vitamin B12</b> or B Complex may be administered with fluids (1 ml per L). Cover fluid bag with a paper bag to prevent degradation of the vitamins.</p> <p>Add KCL to fluids for anorexic cats (20 mEq per liter)</p>                                                                                                                                                                                                                                                                |
| Congestion  | <p>Clean nose with warm moist gauze</p> <p>Humidifier for 10 minutes q 6-24 hrs. (good for foster parents)</p> <p><b>Nebulization</b> for 10 minutes q 6-24 hrs. with sterile saline +/- albuterol, gentamicin, acetylcysteine</p> <p><b>Nose Drops</b> with Gentamicin +/- Dexamethasone – In each nostril BID as needed</p> <p>As a last resort <i>only if the cat is severely congested and not eating</i>, try a decongestant (phenylephrine). One drop BID before feeding in alternating nostrils. Left nostril AM and right nostril PM for no more than 3 days. Continuous use of decongestant nose drops will decrease its effectiveness and cause a rebound effect when stopped.</p> |
| Anorexia    | <p>Correct dehydration, treat congestion, provide analgesia, offer strong-smelling food. Consider appetite stimulants.</p> <p>Perform a complete oral exam. Severe ulceration can cause anorexia and will require analgesia.</p> <p>If ulceration is observed, please check with veterinarian for possible outcome (euthanasia vs quarantine).</p> <p>If steps given above fail and the cat does not eat for &gt; 2 days, administer appetite-stimulating drugs: <b>Mirtazapine TD</b> 0.1ml or 15 mg tablets – 1/8 tablet q 24 hrs. PO (Mirtazapine demonstrates anti-nausea properties as well)</p> <p>-OR-</p> <p>Cyproheptadine 2–4 mg per cat q 12–24 hrs. PO</p>                       |

|  |                                                                                                                                                                                                                                                                 |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>-OR-</p> <p>Midazolam 1.25 mg SQ or IM once with food present</p> <p>If appetite does not respond to appetite stimulants, and anorexia persists for more than 5 days, consider placing an esophageal feeding tube.</p> <p>Monitor weight at least weekly</p> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pain  | <p>Assess pain associated with oral or nasal planum ulceration, corneal ulceration, or severe systemic disease (e.g. joint pain associated with calicivirus, pain associated with dehydration), and treat accordingly. Assess renal function and hydration status prior to administration of NSAIDs.</p> <p><b>Buprenorphine SR</b> (sustained release) 0.12 mg/kg q72 hours SQ</p> <p><b>Buprenorphine</b> 0.01 – 0.02 mg/kg q4-6 hrs. prn IM, SQ, trans-mucosal</p> <p><b>Meloxicam</b> 0.1 mg/kg once PO, SQ. Subsequent dose starting 24 hours after initial dose at 0.05 mg/kg q 24 hrs. PO x 3 days.<sup>4</sup></p> <p><b>Lidocaine</b> 4% viscous. Apply directly to affected area q 6-8 hrs. prn.</p> |
| Fever | <p>If temperature &gt; 106.5 F, administer fluids (subcutaneous or intravenous)</p> <p>Rule-out other systemic problems or lower respiratory tract involvement (thoracic radiographs if possible)</p> <p>Antipyretic drug treatment only if temperature &gt; 106.5 F and fails to respond to external cooling treatment</p>                                                                                                                                                                                                                                                                                                                                                                                    |

## Diagnostic Tests

|                                 |                                                                                                                   |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------|
| PCR Respiratory Pathogen Panel  | Deep pharyngeal and/or nasal swab or conjunctival swab samples needed for submission to a laboratory (e.g. IDEXX) |
| Aerobic Culture and Sensitivity | Deep pharyngeal and/or nasal swab or conjunctival swab samples needed for submission to a laboratory (e.g. IDEXX) |
| Radiographs                     | Minimum of two views of thorax to determine lower respiratory tract involvement                                   |



### 1.2.2 Canine Diarrhea

The following section includes information on assessing, testing, and treatment of canine diarrhea cases within the shelter.

| Assessment                               |                                                                                                                     |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| History                                  | Testing                                                                                                             |
| Duration of illness                      | Fecal – Float                                                                                                       |
| Antibiotic history (past and/or present) | Fecal – Direct (especially in chronic cases)                                                                        |
| Hydration status                         | Parvo SNAP test – perform if blood is present in stool, dog/puppy appears extremely ill, or is found dead in kennel |
| Temperature                              |                                                                                                                     |
| Vomiting +/-                             |                                                                                                                     |

| Treatment                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adults: Non-Complicated; Negative Fecal | <b>Administer SQ fluids</b> to correct dehydration if needed<br><br>Begin treatment with <b>Metronidazole</b> 10-25 mg/kg SID PO for 5 days<br>-AND-<br><b>Panacur</b> PO SID x 3 days<br>-OR-<br>Feed bland diet such as Science Diet I/D for 3-5 days, then gradually reintroduce normal food<br>-AND/OR-<br>Saccharomyces or Fortiflora SID x 5-7 days<br><br>If no improvement, Give 3 days of Ponazuril and repeat 1 dose in 2 weeks even if fecal float is negative |
| Adults: Positive Fecal                  | <b>Roundworms: Strongid</b> 1 ml/10 lbs PO                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | <p><i>Hookworms/Whipworms:</i> <b>Panacur</b> SID PO for 3 days. Repeat in 2-3 weeks</p> <p><i>Coccidia:</i> <b>Ponazuril Paste</b> (Mix tube with 21 cc water) 1 ml/ 5 lbs PO for 3 days</p> <p>- OR-</p> <p><b>Ponazuril</b> (227 mg/ml suspension) - 1 ml/ 10 lbs PO for 3 days</p> <p><i>Giardia:</i> <b>Metronidazole</b> 10-25 mg/kg SID PO for 5 days</p> <p>-OR-</p> <p>Panacur SID PO for 5 days</p> <p><i>Tapeworms:</i> <b>Praziquantel</b> (Droncit) 56.8 mg/mL injectable:</p> <p>&lt;5lbs: 11.4 mg (0.2 mL)</p> <p>5-10 lbs: 22.7 mg (0.4 mL)</p> <p>&gt;10 lbs: 31.4 mg (0.6 mL maximum)</p> |
| Positive Parvo SNAP test                                    | <p>Single dog or puppy - euthanize</p> <p>Puppy with littermates:</p> <p><i>Option 1:</i> euthanize positive puppy and all littermates</p> <p><i>Option 2:</i> euthanize positive puppy → test all littermates with Parvo SNAP test → move negative puppies to available isolation area (400 etc.) → euthanize positive puppies (this option is dependent on the amount of space available in kennel; if there is no space available, consult with the staff veterinarian)</p>                                                                                                                              |
| Chronic or Complicated Cases with Signs of Systemic Disease | Consult staff veterinarian                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### 1.2.3 Canine Skin Lesions

The following section includes information on the most common skin conditions in dogs/puppies, characteristics of these conditions and confirmatory tests. Therapeutic guidelines are also included.

| Things that cause crusting, redness & patchy hair loss |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Skin condition                                         | Characteristics                                                  | Confirmatory Tests                                                                                                                                                                                                                                                                                                                                                                               |
| Dermatophytosis (Ringworm)                             | Contagious and zoonotic *wear gloves*<br><br>Often pruritic      | <b>Positive Wood's lamp test</b><br>- This should NOT be used as a sole test; use in conjunction with skin scrape/fungal culture<br>- Urine, nasal discharge, medications, and fluorescein stain can glow under lamp<br><br><b>Positive DTM (traditional)</b> - media turns red within 10 days<br><br><b>Positive DTM (Rapid Vet-D)</b> - media turns red within 4 days<br><br><b>Fungal PCR</b> |
| Sarcoptic Mange (Scabies)                              | Contagious and zoonotic *wear gloves*<br><br>Severely pruritic   | <b>Positive skin scrape</b><br><br><b>Response to therapy (see below)</b>                                                                                                                                                                                                                                                                                                                        |
| Pyoderma (Bacterial Skin Infection)                    | Non-contagious<br><br>+/- pruritus                               | <b>Skin cytology</b> <ul style="list-style-type: none"> <li>• impression smear</li> <li>• swabs of crusts and exudative skin</li> <li>•</li> </ul> <b>Response to therapy (see below)</b>                                                                                                                                                                                                        |
| Allergic Dermatitis                                    | Non-contagious<br><br>Main Causes: fleas, food allergy, seasonal | <b>Rule out other causes</b> <ul style="list-style-type: none"> <li>• Negative skin scrape</li> <li>• Negative DTM</li> </ul>                                                                                                                                                                                                                                                                    |

|         |                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | <p>History of pruritus (seasonal or nonseasonal)</p> <p>Pruritus that persists beyond resolution of skin infections/antibiotic therapy</p> <p>Affected areas:</p> <ul style="list-style-type: none"> <li>- Tail base &amp; behind the ears<br/>→ suspect flea allergy</li> <li>- Involving ventrum +/- in between toes:<br/>→ suspect food or seasonal allergies</li> </ul> | <p>May see partial or complete response once starting proper <b>flea control</b> or after <b>Cytopoint</b> injection (see below for therapeutic guidelines)</p> |
| Demodex | Common in puppies and immunocompromised dogs                                                                                                                                                                                                                                                                                                                                | <p><b>Positive skin scrape</b></p> <p><b>Response to therapy (see below)</b></p>                                                                                |

| Therapeutic Guidelines     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dermatophytosis (Ringworm) | <p><i>Single spot</i> - <b>Miconazole 1% lotion</b> apply to skin SID or BID for at least 6 weeks based on culture results<br/>DTM culture at weeks 4 &amp; 6; then every 2 weeks as needed<br/>Once 2 negative cultures are obtained 2 weeks apart, treatment can be discontinued</p> <p><i>Generalized, multiple spots:</i><br/><b>Ketoconazole</b> 10 mg/kg PO SID for 14 days<br/>DTM culture at weeks 2 &amp; 4; then every 2 weeks as needed<br/>Once 2 negative cultures are obtained 2 weeks apart, treatment can be discontinued<br/>-OR-<br/><b>Lime Sulfur Dip</b> 2x weekly</p> |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sarcoptic Mange (Scabies)           | <p><b>Selamectin</b> (topical): every 2 weeks; treat for 1-week past resolution of clinical signs</p> <p>&lt;5 lbs: 0.13 ml<br/>&gt;5 lbs: 0.38 ml</p> <p>-OR-</p> <p><b>Ivermectin</b> (injectable) 0.01 mg/lb once weekly; treat for 1-week past resolution of clinical signs</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Pyoderma (Bacterial Skin Infection) | <p>Surface Infection:<br/>Apply <b>2% to 4% chlorhexidine solution</b> topically to affected areas every 12 hours for 1-2 days past resolution of clinical signs<br/>( + topical ketoconazole if yeast present) - may use medicated shampoos</p> <p>Superficial Pyoderma:<br/><i>*Treat for <u>at least 2 weeks, and 1-week past resolution of clinical signs</u></i><br/><b>Medicated baths once weekly</b></p> <p>-AND-</p> <p><b>Cefpodoxime</b> 5-10 mg/kg PO Q24h</p> <p>-OR-</p> <p><b>Cephalexin</b> 22-30 mg/kg PO Q8-12h</p> <p>-OR-</p> <p><b>Clavamox</b> 20 mg/kg PO Q8-12h</p> <p>Deep Pyoderma:<br/><i>*Treat for <u>at least 4 weeks, and 1-week past resolution of clinical signs</u></i><br/><b>Medicated baths once weekly</b></p> <p>-AND-</p> <p><b>Cefpodoxime</b> 5-10 mg/kg PO Q24h</p> <p>-OR-</p> <p><b>Cephalexin</b> 22-30 mg/kg PO Q8-12h</p> <p>-OR-</p> <p><b>Clindamycin</b> 11 mg/kg PO Q12h</p> <p><i>*May be recommended when draining tracts, fistulas, or anal sac infections are also present</i></p> <p>-OR-</p> <p><b>Clavamox</b> 20 mg/kg PO Q8-12h</p> |
| Allergic Dermatitis                 | <p>Start on appropriate flea preventative - <b>Selamectin</b></p> <p>For anti-inflammatory/ antipruritic effects:<br/><b>Cytopoint</b> at a minimum dose of 2mg/kg</p> <p>-OR-</p> <p><b>Prednisolone</b> start with 1 mg/kg PO BID for 3 days → then SID for 3 days → then SID every other day for 3 more doses</p> <p><i>*See treatment guidelines for pyoderma if secondary skin infections are present</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Demodex                             | <p><b>Bravecto</b> for dogs- Each chew is formulated to provide a minimum dose of 11.4 mg/lb (25 mg/kg) body weight every 3 weeks</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

*These are the most common canine skin lesions seen in a shelter setting. Other non-wound skin lesions such as tumors, etc. should be discussed with the staff veterinarian on a case-by-case basis*

#### 1.2.4 Canine Wound Care

- Assess wounds for depth, chronicity, and signs of infection
- Procedural sedation may be necessary to assess and treat wounds
- Clip and clean wounds whenever possible
- If maggots present: a Capstar tablet can be crushed and sprinkled on the wound bed. This will paralyze the maggots and make cleaning the wound easier.
- Superficial wounds can often be debrided with gauze then managed with **SSD, honey, or Zn7** topically BID until new skin has healed
- Flush deeper wounds (puncture holes, abscesses, etc.) with **diluted chlorhexidine solution**
- Some deeper wounds or abscess pockets may require surgery and placement of a drain by the staff veterinarian. Drains are usually left in place until the wound stops draining.
- For deep wounds, it is recommended to start antibiotic therapy to treat or prevent infections. Treatment duration is typically between 10 and 14 days, however it is recommended to continue antibiotic therapy for at least 2-3 days or 1-week past resolution of clinical signs.

Preferred antibiotics for canine wounds include:

- o **Convenia** - 3.6 mg/lb SQ q14 days (small dogs)
  - o **Clindamycin**
    - a. Capsules: 25 mg/ 10 lb BID
    - b. Liquid: 5 mg/lb SID
  - 3. **Enrofloxacin (Baytril)**
    - a. 5-20 mg/kg per day PO (SID or BID)
    - b. Young, pregnant, or suspect renal disease: 2.5 - 5 mg/kg PO or IM q12h
    - c. Maximum duration of treatment: 30 days
- 
- If wound infections fail to resolve or worsen after starting antibiotic therapy, it may be necessary to switch or add antibiotics to the treatment protocol. In rare cases, a culture and sensitivity may be indicated if the wound is not responding to any of the treatments listed above. Please consult with a veterinarian.

#### 1.2.5 Bite Wounds (New or Infected)

- Irrigation and debridement should be performed if possible
- Culture should be considered if no response to antibiotics after 3 days

Preferred antibiotics for bite wounds include:

- o Doxycycline 10mg/kg PO SID X 5 days
- o Amoxicillin-clavulanate (Clavamox) 13.75-20mg/kg PO BID X 5 days

### 1.2.6 Canine Diarrhea – Extended Treatment & Escalation Protocol

This protocol supplements 1.2.2 Canine Diarrhea, providing a step-by-step escalation pathway to follow when a dog's fecal is negative and diarrhea does not resolve with the initial treatment in 1.2.2.

#### 1. Initial Assessment

1. Observe and document hydration status, appetite, energy level, and stool characteristics (consistency, frequency, presence of blood, mucus, or parasites).
2. If diarrhea is bloody, perform a parvovirus antigen test immediately, regardless of age, before any other treatment or testing.

#### 2. Diagnostic Testing

Perform a fecal flotation (or equivalent fecal exam) on all dogs with diarrhea and record results in the medical record. If positive, treat based on the identified parasite/pathogen per 1.2.2.

#### 3. Step-by-Step Treatment (if fecal is negative)

1. **Fenbendazole (Panacur)** – once daily for 3 days, dosed per weight chart.
2. **Pro-Pectalin** – 1 chewable tablet per 20 lbs body weight, twice daily for 2 days.
3. **FortiFlora** – 1 packet mixed with food once daily until diarrhea resolves, then continue 1 week afterward. May be given with or instead of Pro-Pectalin.
4. **Important:** Pro-Pectalin and FortiFlora must not be given at the same time as antibiotics – separate doses by at least 1–2 hours. Special/GI-support diets may be started at the veterinarian's discretion.
5. **If diarrhea continues: Metronidazole** 10–15 mg/kg PO BID for 5–7 days.
6. **If diarrhea persists after Metronidazole:** repeat the fecal flotation before any further testing.
7. **If the repeat fecal is still negative and diarrhea persists:** submit a fecal PCR panel for further diagnostics.

#### 4. Reevaluation and Monitoring

1. Monitor and document stool consistency daily.
2. Reassess hydration, weight, and overall condition throughout treatment.
3. Escalate to the veterinarian if diarrhea worsens, fails to improve, or systemic illness develops.

#### Documentation

Record all test results, treatments, diet changes, and progress notes in the medical record, with medication dosages and durations clearly documented.

#### Notes

Supportive care (fluids, anti-nausea medications, etc.) may be provided at the veterinarian's discretion. Dogs with confirmed or suspected contagious causes must also follow the SOP specific to that pathogen (e.g. Parvovirus) in addition to this protocol.

### 1.3 Perioperative (Unaltered) Procedures

|                            | Dogs/Puppies                                                                                                                                                                                                                                                                                                                                                                                                    | Cats/Kittens                                                                                                                                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Analgesics</b>          | <p><b>Preoperative</b> -Rimadyl SQ at 2.2mg/kg</p> <p><b>Post-operative</b> – Rimadyl PO 2.2mg/kg SID x 3 days<br/>-OR-<br/>Previcox 5mg/kg SID x 3 days<br/>-AND/OR-<br/>Buprenorphine SR 0.03-.06 mg/kg q 72 hours SQ if directed by veterinarian</p>                                                                                                                                                         | <p><b>Preoperative</b> -Metacam SQ at 0.2mg/kg</p> <p><b>Post-operative</b> – Buprenorphine SR 0.12mg/kg q 72 hours SQ if directed by veterinarian<br/>Meloxicam starting 24 hours after initial dose at 0.05 mg/kg q 24 hrs. PO x 3 days if directed by veterinarian</p>  |
| <b>Antibiotics</b>         | Prescribed only if directed by the veterinarian                                                                                                                                                                                                                                                                                                                                                                 | Prescribed only if directed by the veterinarian                                                                                                                                                                                                                            |
| <b>Heartworm Treatment</b> | <p>During Surgery: All positive dogs will be treated with Immiticide or <b>Diroban</b> per packaging dosing after a 2-week course of Doxycycline.</p> <p><b>2 treatments 24 hours apart</b> unless 3 treatment plan is recommended (1 treatment followed by 2 treatments 24 hours apart one month after the initial treatment)</p> <p>Doxycycline will be continued for an additional 2 weeks post-surgery.</p> |                                                                                                                                                                                                                                                                            |
| <b>Other Medications</b>   | <p><b>Cerenia</b> – Given preoperatively to all dogs/puppies at a dose of 1-2mg/kg SQ every 24 hours</p> <p><b>Heartworm + Dogs:</b> All positive dogs will be treated with Immiticide or <b>Diroban</b> per package dosing at time of surgery. <b>2 treatments 24 hours apart</b> unless 3 treatment plan is</p>                                                                                               | <p><b>Cerenia</b> – Given preoperatively to cat/kittens at a dose of 1mg/kg SQ every 24 hours</p> <p><b>Praziquantal:</b> All cat/kittens. Give in accordance with formulary</p> <p><b>Flea/Heartworm Control:</b> Apply Revolution (if not previously treated) at the</p> |



|                      |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                          |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | <p>recommended by veterinarian (1 treatment followed by 2 treatments 24 hours apart one month after the initial treatment)</p> <p><b>Flea/Heartworm Control:</b> Apply Revolution (if not previously treated) at the rate of 6mg/kg (~ ¼ ml per 10 lbs)</p> <p><b>Tapeworms:</b> If seeing tapeworms in stool, treat with Praziquantal (Droncit) in accordance with formulary</p> | <p>rate of 6mg/kg (~ ¼ ml per 10 lbs)</p> <p><b><u>Day following surgery:</u></b></p> <p><b>Ponazuril</b> - kittens under 6 months. 1cc/5 lbs</p> <p><i>Strongid and Ponazuril are prepared and placed on cage to be given the following morning</i></p> |
| <b>Vaccinations</b>  | <p>Ensure dogs are <b>current</b> on <b>Rabies, DHPP, and Bordetella</b> vaccination.</p> <p><i>Initial vaccination should have been completed at intake. See Intake Protocol</i></p>                                                                                                                                                                                             | <p>Ensure cats/kittens are current on <b>Rabies</b> and <b>FVRCP</b> vaccination.</p> <p><i>Initial vaccination should have been completed at intake. See intake Protocol</i></p>                                                                        |
| <b>Microchipping</b> | <p>All adoption dogs/puppies and reclaims (if owner wants a chip) needing surgery are chipped at the time of surgery.</p>                                                                                                                                                                                                                                                         | <p>All adoption cats/kittens needing surgery are chipped at the time of surgery.</p>                                                                                                                                                                     |
| <b>Miscellaneous</b> | <p><b>Ear Exam:</b> Examine the ears of all animals following surgery and clean as appropriate.</p> <p><b>Nail Trim:</b> Examine the nails of all animals following surgery and trim as appropriate</p>                                                                                                                                                                           | <p><b>Ear, Nail, and Flea Control</b> – Same as Dog/Puppies</p>                                                                                                                                                                                          |

## 1.5 Post-Operative Examinations

Post-operative examinations are provided to adopters within seven days of adoption. Animals needing care after the 7 days window should be discussed with the veterinarian prior to scheduling or denying treatment. Other surgeries needing post-operative examination should also be discussed with the veterinarian.

### 1.5.1 Post-OHE Examination

- **Swelling:** A certain degree of swelling is normal after surgery. Typically, this is caused by a reaction to the suture material buried in the surgical site. A hard swelling of “finger” thickness or less with no signs of drainage or opening of the skin is generally not a problem. If the owner wishes, they can place warm compresses on the surgical wound two to three times daily to promote circulation and removal of the swelling. Cats are generally more sensitive to the suture material than dogs, especially if SQ sutures were required during surgery. Swelling greater than “finger” thickness should be brought to the attention of a veterinarian.
- **Redness with no drainage,** excessive swelling or wound disruption: Inflammation of the surgical wound may be due to a number of things. Often it indicates that the animal has been licking its incision site. Be sure to thoroughly question the owner on this. An Elizabethan collar is in order if this is the case.
- **Redness accompanied by drainage, swelling or wound disruption** – inform a veterinarian.

### 1.5.2 Post-Castration Examination

- **Scrotal Swelling** – Typically this is more common in dogs and is often the result of a scrotal hematoma. These hematomas can vary in size. A good rule of thumb is that if the animal doesn’t look like it has been castrated (swelling about the size of its former testicles) it’s probably ok – have the owner observe for further changes and ensure the animal is following post-operative restrictions on activity and is wearing an E-collar. If the swelling is greater, bring the condition to the attention of a veterinarian.
- **Redness with no drainage,** excessive swelling or wound disruption: Inflammation of the surgical wound may be due to a number of things. Often it indicates that the animal has been licking its incision site. Be sure to thoroughly question the owner on this. An Elizabethan collar is in order if this is the case.
- **Redness accompanied by drainage, swelling or wound disruption** – inform a veterinarian.

## **1.6 Owned Animal Services**

Pinellas County Animal Services provides Rabies vaccinations and microchips to privately owned animals at the time of reclaim or during scheduled weekly rabies clinics. All services must include a signed consent form from the owner provided by customer service representatives at the time of service.

### **1.6.1 Vaccination**

Animal Services provides rabies vaccinations for dogs and cats over 12 weeks of age. Initial vaccination – as long as the animal is 12 weeks or older – must be followed by a booster in one year. Subsequent vaccinations (as long as they are given at least one year after the initial vaccination) will have duration of immunity consistent with the manufacturers' recommendation (currently 3 years for the vaccine use by PCAS). Animal Services does not vaccinate the public's animals for diseases other than Rabies nor do our personnel recommend non-veterinary sources of vaccine. Rabies vaccines must be given by a veterinarian.

### **1.6.2 Microchipping**

Animal Services provides microchipping for both dogs and cats. All animals should be scanned prior to microchipping to ensure the animal does not already have a microchip. All microchips should be inserted as directly by the manufacturer. Once the microchip is inserted, the animal should be rescanned to ensure the microchip is working.

## **1.7 Records Maintenance**

All exams and medical care will be placed in PetPoint under the animals ID. These records will remain on record for a minimum of three years.

## 2 Controlled Substance Management

**Purpose:** To provide basic guidelines for the ordering, receiving, securing, utilization, and inventory of controlled substances used by authorized individuals of Pinellas County Animal Services (PCAS).

**Scope:** This SOP applies to all PCAS employees who order, receive, utilize, or for any reason have access to controlled substances.

**References:** US Department of Justice, Drug Enforcement Administration, Practitioner's Manual, 2006 Edition

### 2.1 Locations/Contents

There are several numbered safes and areas located in the kitten nursery area in the main shelter and the medical building which contain various logs, manuals, and controlled substance inventories. The following list will indicate the contents of each safe.

#### 2.1.1 Logs - Cabinet in Kitten Nursery (Main Shelter)

Logs/Binders

1. Inspections and Invoices (CIII, IV, V)
2. Fatal Plus Invoices
3. Transfer Log (CIII, IV, V)
4. Fatal Plus Transfer Log (CII)
5. Fatal Plus Weekly Records Review
6. Fatal Plus log – exams
7. TX Log -exams

#### 2.1.2 SAFE #3 – Floor Safe (Main Shelter)

Main Inventory

1. Fatal Plus
2. Box for expired drugs (Labeled “EXPIRED”)

#### 2.1.3 SAFE #4 – E-safe (Main Shelter)

Inventory

1. Fatal Plus (CII) – from main inventory (safe #3)
2. TX – from main inventory (transfer from safe #2 -Medical Building)

#### **2.1.4 SAFE #1– Medical Building - In Lab Area**

Main Inventory

1. C III, IV, V
2. Box for expired drugs (Labeled “EXPIRED”)

Surgery Drug Log (CIII, IV, V) is located next to safe in locked cabinet

#### **2.1.6 SAFE #2 – Tech Safe – Transfer Safe (Medical Building)**

Inventory

1. C III, IV, V
2. Box for expired drugs (Labeled “EXPIRED”)

Surgery Drug Log (CIII, IV, V) is located next to safe in locked cabinet

### **2.2 Staff Access**

A list of staff with safe access is located in Safe #1 in the Controlled Substance Handbook. This list should be updated every 4-6 months to ensure it is up to date.

### **2.3 Procedures**

The following is a list of controlled substances and logs by procedures ( i.e., surgery, dog/cat exams, prepay, wildlife, bite dogs)

#### **2.3.1 Surgery – CVTs**

1. Use Surgery Drug Log (CIII, IV, V) located next to safe #2 in locked cabinet
2. Use drugs needed for surgery located in safe #2
3. Enter date, animal ID, amount of drug(s) used, new total amount in bottle, and initials in surgery drug log

#### **2.3.2 Dog/Cat Exams and Prepay by CVTs**

1. Use Fatal Plus Log and TX log from locked cabinet in kitten nursery
2. Use Fatal Plus from safe #4 – E-safe
3. Enter date, animal ID, amount of drug(s) used, new total amount in bottle, and initials in Fatal Plus and TX logs

#### **2.3.3 Prepay by ACAs**

4. Use Fatal Plus Log and TX log from locked cabinet in kitten nursery
1. Use Fatal Plus from safe #4 – E-safe

2. Enter date, animal ID, amount of drug(s) used, new total amount in bottle, and initials in Fatal Plus and TX logs

#### **2.3.4 EUs by ACOs**

5. Use Fatal Plus Log and TX log from locked cabinet in kitten nursery
1. Use Fatal Plus from safe #4 – E-safe
2. Enter date, animal ID, amount of drug(s) used, new total amount in bottle, and initials in Fatal Plus and TX logs

## **2.4 Ordering**

All orders for controlled substances are placed through finance. For Fatal Plus or other CII substance, a 222 Form must be filled out and signed by the form's responsible representative. If ordering Fatal Plus, please have Rusty Walker sign the form before ordering.

The following steps should be followed for all CII substances.

1. Once form is signed, remove back copy of 222 form. This back copy is placed in the Fatal Plus Invoice binder. The original back copy stays in medical until product arrives.
2. The remaining two copies of the 222 form are given to finance to mail to the distributor when order is placed.
3. After product is received and counted, 2 employees (mangers, finance etc), will initial the original 222 to verify that the order is correct.
4. 2 employees (mangers, finance etc), will initial and date the invoice received
5. A copy of the original 222 form and the invoice will be made. The stapled originals stay with finance and the stapled copies are placed back in the Fatal Plus Invoice binder located in the locked cabinet in the nursery.
6. New inventory is recorded in the Fatal Plus Transfer Log (nursery cabinet) along with 2 initials 2 employees (mangers, finance etc) The new inventory is stocked in safe #3.

All other controlled substance (CIII, IV, V) orders are also placed through finance. A copy of the invoices is kept next to Safe #1 in the Inspections and Invoices binder. New inventory is stocked in Safe #1 and is logged in the Transfer Log (CIII, IV, V) with the corresponding invoice number. Both the invoice and the transfer log entry should be initialed by the employee.

## **2.5 Transferring Stock**

All new stock transferred from safe #1 and logged as new stock in the logbooks must also have the expiration date written in the log next to the substance being used. This will help to ensure that stock used is within the expiration date noted on the bottle.

## 2.6 Inventories/Record Keeping

### 2.6.1 Daily

All entries entered in control logs (surgeries, prepays, etc.) are to be inventoried daily to check for entry errors or mathematical errors.

0.05ml – 0.1ml per entry is considered a reasonable range of hub loss and can be considered waste. For example, if there are 20 entries then hub loss can be determined by multiplying 0.05ml – 0.1ml X 20. Therefore, hub loss would be 1 ml to 2 ml.

### 2.6.2 Weekly

A Fatal Plus weekly records review is completed weekly. Entries in the Fatal Plus Log are compared to the number of euthanasias entered into PetPoint during the same time period. Information obtained should be entered into the Fatal Plus Weekly Records Review located in the locked cabinet in the kitten nursery. The following should be entered:

1. Date Time Span – Should include date after last review to present time.
2. Total # of PetPoint euthanasia entries during time span
3. Total # of logbook euthanasia entries from both Fatal Plus Logs (combined total)
4. Number of discrepancies found
5. Number of discrepancies corrected
6. Initials of employee during the review
7. Date completed

**In PetPoint** – The report is Animal: Outcome. Choose an outcome start date and an outcome end date. Choose Euthanasia for outcome type. The report should give you Total Outcomes or euthanasias. You can compare this number to the total number of entries from both Fatal Plus logs combined

### 2.6.3 Biennial

Required by the DEA. The DEA requires a physical inventory of all controlled substance to be conducted **every two years**. The inventory may be taken on any date within two years of the previous inventory date. This inventory must be kept at the registered site for two years after it is taken. There are two forms for Biennial Inventory. One is for Schedule II (Fatal Plus) and one is for Schedule III, IV, V. These forms can be found on SharePoint.

---

## 2.7 Expired Drug Disposal

All “**expired**” controlled drugs must be maintained separately from other inventory until its destruction or until it is sent to a reverse distributor. Boxes labeled “expired” are located in Safe #2 and Safe #5. All expired controlled drugs should be placed in these boxes.

Form 41 shall be used to record the **destruction of all controlled substances, including controlled substances from acquired collectors.**

- The Form 41 shall include the names and signatures of the **two employees** who witnessed the destruction. Does not have to be sent to DEA but must be kept for 2 years.
  - **Exceptions for DEA Form 41:**
3. Transfers by registrant to a reverse distributor must be recorded in accordance with 1304.22 (c), and such record **need not** be maintained on a Form 41
  3. Destruction of a controlled substance dispensed by a veterinarian for immediate administration at the veterinarians registered location, when the substance is not fully exhausted (i.e. wastage) shall be properly recorded in accordance with 1304.22 (c), and such record **need not** be maintained on a Form 41

**Summary:** If controlled substances are disposed of due to spillage, broken bottle etc., (not hub loss) Form 41 should be used with two employee signatures. All other controlled drugs will be transferred to a reverse distributor.

A printable Form 41 can be found here:

[https://www.deadiversion.usdoj.gov/21cfr\\_reports/surrend/41\\_form.pdf](https://www.deadiversion.usdoj.gov/21cfr_reports/surrend/41_form.pdf)

A printable reverse distributor form can be found

here: [https://returnlogistics.com/wp-content/uploads/2018/05/RLI\\_ASN.pdf](https://returnlogistics.com/wp-content/uploads/2018/05/RLI_ASN.pdf)

**Noncontrolled expired medications** can be disposed of either by reverse distributor or by mixing medications (liquid or pills) with an unappealing substance such as cat litter or dish soap and placed in trash

## 2.8 Theft/Loss Reporting

Federal regulations require that registrants notify the Field Division Office of the Administration in his area, in writing, of the theft or significant loss of any controlled substance, disposal receptacles or listed chemicals **within one business** day of discovery of such loss or theft. The registrant shall also complete and submit to the Field Division Office in his area, DEA Form 106 regarding the loss or theft (21 C.F.R. §1301.76(b) and 21 U.S.C. §830(b)(1)(C)). The DEA Form 106 can be completed via **Theft/Loss Reporting Online (TLR)** or download the **fillable PDF** version and submit to your **Local Diversion Field Office**.



DEA Form 106 Online: [https://www.deadiversion.usdoj.gov/21cfr\\_reports/theft/index.html](https://www.deadiversion.usdoj.gov/21cfr_reports/theft/index.html)

DEA 106 Printable Form: [https://www.deadiversion.usdoj.gov/21cfr\\_reports/theft/DEA\\_Form\\_106.pdf](https://www.deadiversion.usdoj.gov/21cfr_reports/theft/DEA_Form_106.pdf)

### **Tampa DEA Field Division Office**

4950 W. Kennedy Blvd, Suite

400Tampa, FL 33609Diversion Number: (813) 287-4765

Diversion Fax: (813) 287-4766

*Jurisdiction:* West Central Florida

## **2.9 Sample Forms**

Needed forms (Biennial Controlled Substance Inventory, Record of Controlled Substances Destroyed, Report of Theft or Loss etc.) can be found in the Inspections and Invoices binder located in Safe #1